

Class 4: Simple Harmonic Motion (Part 2)

Summer Break Bridge Course

Recap: Standard Form of SHM

In the previous class, we obtained the general displacement of a particle undergoing simple harmonic motion in the form

$$x(t) = A \sin(\omega t + \alpha),$$

where:

- A is the amplitude,
- ω is the angular frequency,
- α is the initial phase or phase constant.

If we define

$$\theta(t) = \omega t + \alpha,$$

then the displacement may be written as

$$x(t) = A \sin \theta(t).$$

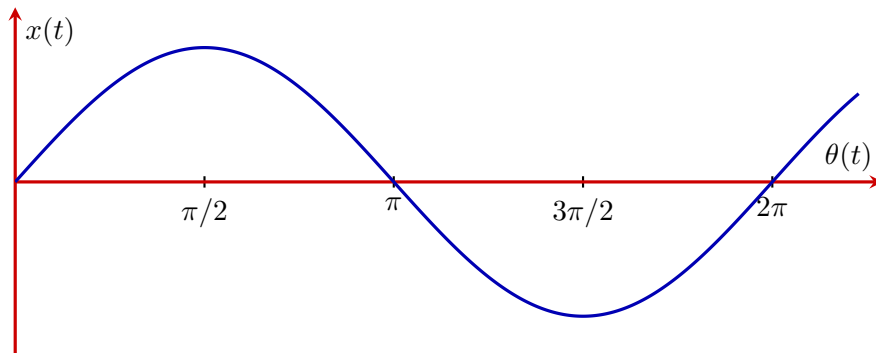
Time Period and Frequency

The **time period** T is the time needed to complete one full oscillation.

For SHM,

$$x(t) = A \sin(\omega t + \alpha). \tag{1}$$

A full cycle means that the phase increases by 2π because the sin-function has a period of 2π .



Therefore, if T is the time period, then

$$x(t + T) = x(t) \tag{2a}$$

$$A \sin(\omega(t + T) + \alpha) = A \sin(\omega t + \alpha). \tag{2b}$$

For the sine function to repeat its value, we need

$$\begin{aligned}\omega(t + T) + \alpha &= \omega t + \alpha + 2\pi \\ \omega T &= 2\pi \\ T &= \frac{2\pi}{\omega}\end{aligned}\tag{3}$$

Frequency

The **frequency** ν is the number of complete oscillations per unit time. Therefore,

$$\boxed{\nu = \frac{1}{T} = \frac{\omega}{2\pi}}.\tag{4}$$

So we also get

$$\boxed{\omega = 2\pi\nu}.\tag{5}$$

The quantity ω in SHM is called the **angular frequency**. To understand why, it is helpful to compare SHM with uniform circular motion.

In uniform circular motion, the angular velocity tells us how much angle is swept per unit time. If a particle completes one full revolution, then the angular displacement is 2π radians. So, if it makes ν complete revolutions per second, its angular velocity is

$$\omega = 2\pi\nu.$$

Now in SHM, the particle is not literally going around a circle, but its motion can still be described by a **phase**

$$\theta(t) = \omega t + \alpha.$$

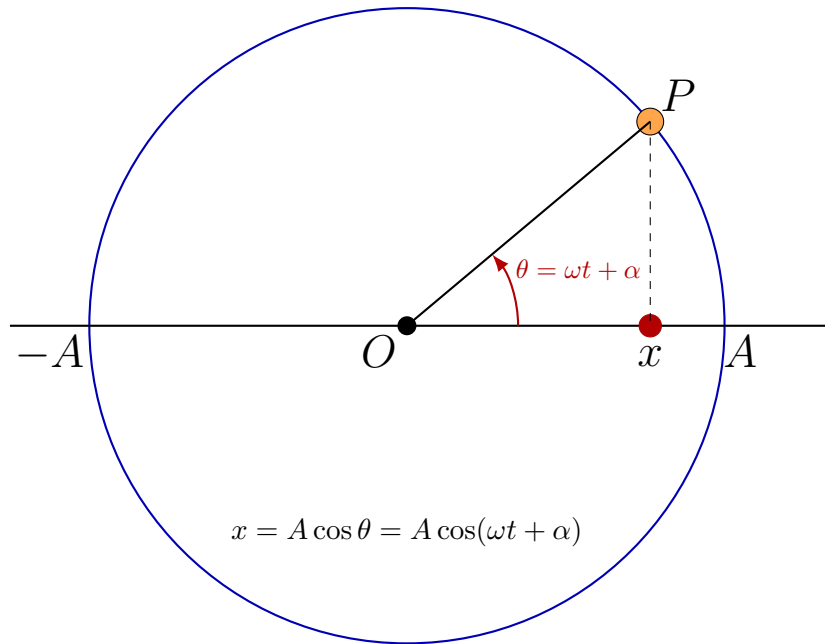
A complete oscillation corresponds to an increase of this phase by 2π , just like one full revolution in circular motion corresponds to an angular change of 2π .

So, in SHM, ω tells us how fast the *phase* is changing with time, just as angular velocity tells us how fast the angle changes with time in circular motion. That is why ω plays the role of an angular-velocity analogue in SHM.

SHM as a Projection of Uniform Circular Motion

A very useful way to visualize SHM is to imagine a point moving with constant angular speed ω on a circle of radius A . If we look only at the projection of that point onto a horizontal diameter, then that projection executes simple harmonic motion.

$$\text{uniform circular motion} \implies \text{projection on a diameter gives SHM}$$



In this picture:

- the point P moves uniformly on the circle,
- the angle θ increases uniformly with time as

$$\theta = \omega t + \alpha,$$

- and the projection x of that point on the horizontal diameter moves back and forth between $-A$ and A .

So the amplitude of SHM is the radius A of the circle, and the angular frequency of SHM is the same as the angular speed of the rotating point.

This analogy is extremely important:

- in circular motion, ω is the angular velocity,
- in SHM, ω is the rate of change of phase,
- and SHM can be viewed as the shadow or projection of uniform circular motion on a diameter.

Phase and Phase Difference

For

$$x(t) = A \sin(\omega t + \alpha) = A \sin \theta(t),$$

the quantity

$$\theta(t) = \omega t + \alpha$$

is called the **phase of oscillation**.

At $t = 0$,

$$\theta(0) = \alpha.$$

The initial phase α is sometimes called the **epoch**.

Phase Difference

If two SHMs are given by

$$x_1(t) = A_1 \sin \theta_1(t), \quad x_2(t) = A_2 \sin \theta_2(t),$$

then the phase difference at time t is

$$\boxed{\theta_1(t) - \theta_2(t)}.$$

At the initial time $t = 0$, the phase difference is

$$\theta_1(0) - \theta_2(0).$$

Position and Velocity are Out of Phase

For SHM,

$$x(t) = A \sin(\omega t + \alpha),$$

and

$$v(t) = \frac{dx}{dt} = A\omega \cos(\omega t + \alpha).$$

Using

$$\cos \phi = \sin \left(\phi + \frac{\pi}{2} \right),$$

we may write

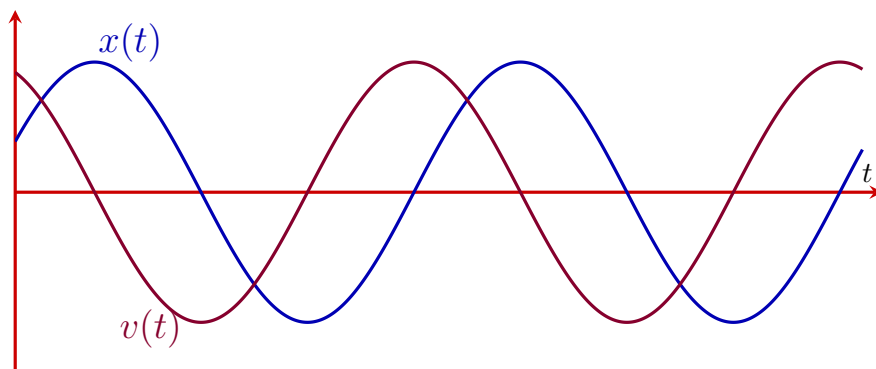
$$v(t) = A\omega \sin \left(\omega t + \alpha + \frac{\pi}{2} \right).$$

Therefore, displacement and velocity differ in phase by

$$\boxed{\frac{\pi}{2}}$$

This means:

- when x is maximum or minimum, $v = 0$,
- when $x = 0$, $|v|$ is maximum.



Some Numerical Problems

Problem 1

A particle undergoing SHM has displacement

$$x(t) = 3 \sin(60\pi t).$$

Find:

- (i) amplitude,
- (ii) time period,
- (iii) acceleration at maximum displacement.

Solution

Comparing with

$$x(t) = A \sin(\omega t + \alpha),$$

we identify

$$A = 3, \quad \omega = 60\pi, \quad \alpha = 0.$$

So,

$$\boxed{A = 3.}$$

The time period is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{60\pi} = \frac{1}{30}.$$

Hence,

$$\boxed{T = \frac{1}{30}.$$

Now acceleration is

$$a = -\omega^2 x.$$

At maximum displacement, $x = A = 3$, so

$$a = -(60\pi)^2 \times 3.$$

Thus,

$$\boxed{a_{\text{at max disp.}} = -10800\pi^2.}$$

Problem 2

If

$$x(t) = 4(\sin \pi t + \cos \pi t) \text{ cm},$$

find the amplitude.

Solution

Here,

$$\begin{aligned}x(t) &= 4 [\sin(\pi t) + \cos(\pi t)] \text{ cm} \\&= 4\sqrt{2} \left[\frac{1}{\sqrt{2}} \sin(\pi t) + \frac{1}{\sqrt{2}} \cos(\pi t) \right] \text{ cm} \\&= 4\sqrt{2} \left[\cos\left(\frac{\pi}{4}\right) \sin(\pi t) + \sin\left(\frac{\pi}{4}\right) \cos(\pi t) \right] \text{ cm} \\&= 4\sqrt{2} \sin\left(\pi t + \frac{\pi}{4}\right) \text{ cm}\end{aligned}$$

Hence, the amplitude is

$$\boxed{4\sqrt{2} \text{ cm.}}$$

Problem 3

The displacement of an SHO is

$$x(t) = 10 \sin\left(\frac{\pi}{3}t - \frac{\pi}{2}\right) \text{ cm.}$$

Find:

- (i) amplitude,
- (ii) time period,
- (iii) highest velocity,
- (iv) highest acceleration,
- (v) initial phase,
- (vi) velocity at $t = 1$ s.

Solution

Comparing with

$$x(t) = A \sin(\omega t + \alpha),$$

we get

$$A = 10 \text{ cm}, \quad \omega = \frac{\pi}{3} \text{ rad s}^{-1}, \quad \alpha = -\frac{\pi}{2}.$$

So,

$$\boxed{A = 10 \text{ cm.}}$$

The time period is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi/3} = 6 \text{ s.}$$

Hence,

$$\boxed{T = 6 \text{ s.}}$$

The maximum speed is

$$v_{\max} = \max (\pm \omega \sqrt{A^2 - x^2}) = A\omega = 10 \cdot \frac{\pi}{3} = \frac{10\pi}{3} \text{ cm s}^{-1}.$$

Thus,

$$v_{\max} = \frac{10\pi}{3} \text{ cm s}^{-1}.$$

The maximum magnitude of acceleration is

$$a_{\max} = \max (-\omega^2 x) = \omega^2 A = \left(\frac{\pi}{3}\right)^2 \times 10 = \frac{10\pi^2}{9} \text{ cm s}^{-2}.$$

So,

$$a_{\max} = \frac{10\pi^2}{9} \text{ cm s}^{-2}.$$

The initial phase is simply

$$\alpha = -\frac{\pi}{2}.$$

Now

$$\begin{aligned} v(t) &= \frac{dx}{dt} \\ &= A\omega \cos(\omega t + \alpha) \\ &= 10 \cdot \frac{\pi}{3} \cos\left(\frac{\pi}{3}t - \frac{\pi}{2}\right). \end{aligned}$$

At $t = 1$,

$$v(1) = \frac{10\pi}{3} \cos\left(\frac{\pi}{3} - \frac{\pi}{2}\right) = \frac{10\pi}{3} \cos\left(-\frac{\pi}{6}\right) = \frac{10\pi}{3} \cdot \frac{\sqrt{3}}{2}.$$

Hence,

$$v(1) = \frac{5\pi\sqrt{3}}{3} \text{ cm s}^{-1}.$$

Problem 4

A particle undergoing SHM completes 60 oscillations in 2 s. Initially it is 5 cm away from equilibrium and has zero velocity. Find its equation of displacement.

Solution

The frequency is

$$\nu = \frac{60}{2} = 30 \text{ s}^{-1}.$$

So

$$\omega = 2\pi\nu = 60\pi \text{ rad s}^{-1}.$$

Let

$$x(t) = A \sin(60\pi t + \alpha).$$

Then

$$v(t) = A(60\pi) \cos(60\pi t + \alpha).$$

At $t = 0$,

$$x(0) = A \sin \alpha = 5,$$

and

$$v(0) = A(60\pi) \cos \alpha = 0.$$

Thus,

$$\cos \alpha = 0.$$

One convenient choice is

$$\alpha = \frac{\pi}{2}.$$

Then

$$A \sin \left(\frac{\pi}{2} \right) = A = 5.$$

So

$$x(t) = 5 \sin \left(60\pi t + \frac{\pi}{2} \right).$$

Equivalently, one may write

$$x(t) = 5 \cos(60\pi t).$$

Problem 5

The frequency of oscillation of a wire is 200 Hz. A point on the wire has velocity 4.35 m s^{-1} when it reaches half its amplitude. Find the acceleration of that point at that instant.

Solution

The angular frequency is

$$\omega = 2\pi\nu = 2\pi \times 200 = 400\pi \text{ rad s}^{-1}.$$

At half the amplitude,

$$x = \frac{A}{2}.$$

For SHM,

$$v = \pm\omega\sqrt{A^2 - x^2}.$$

So at $x = A/2$,

$$v = \omega\sqrt{A^2 - \frac{A^2}{4}} = \omega\sqrt{\frac{3A^2}{4}} = \frac{\sqrt{3}}{2}\omega A.$$

Given $v = 4.35$,

$$\frac{\sqrt{3}}{2}\omega A = 4.35.$$

Hence,

$$A = \frac{2 \times 4.35}{\sqrt{3}\omega}.$$

Acceleration at half amplitude is

$$a = -\omega^2 x = -\omega^2 \frac{A}{2}.$$

Substituting the expression for A ,

$$a = -\omega^2 \cdot \frac{1}{2} \cdot \frac{2 \times 4.35}{\sqrt{3} \omega} = -\frac{4.35 \omega}{\sqrt{3}}.$$

Using $\omega = 400\pi$,

$$a = -\frac{4.35 \times 400\pi}{\sqrt{3}}.$$

Thus,

$$a = -\frac{1740\pi}{\sqrt{3}} \text{ m s}^{-2}.$$

Problem 6

A particle undergoing SHM has positions 6 cm, 10 cm, and 6 cm at three consecutive seconds. Determine its frequency.

Solution Idea

Let

$$x(t) = A \sin(\omega t + \alpha).$$

Take the three times as t , $t + 1$, $t + 2$. Then

$$A \sin(\omega t + \alpha) = 6,$$

$$A \sin(\omega t + \omega + \alpha) = 10,$$

$$A \sin(\omega t + 2\omega + \alpha) = 6.$$

A direct algebraic route is possible, but there is also a useful trick that simplifies the work substantially.

An Important Trick: Shifting the Time Origin

Suppose

$$x(t) = A \sin(\omega t + \alpha).$$

Now define a shifted time coordinate

$$t' = t + \beta.$$

Then

$$x(t) = A \sin(\omega(t' - \beta) + \alpha) = A \sin(\omega t' + \alpha - \omega\beta).$$

So the same SHM can be written as

$$x(t') = A \sin(\omega t' + \alpha'),$$

where

$$\alpha' = \alpha - \omega\beta.$$

Shifting the time origin changes only the initial phase. The amplitude and angular frequency remain unchanged.

For the last numerical problem, this means we are free to choose the three consecutive times as

$$t = 0, 1, 2$$

without changing the frequency. We may then write

$$6 = A \sin \alpha,$$

$$10 = A \sin(\omega + \alpha),$$

$$6 = A \sin(2\omega + \alpha).$$

This reduces the algebra to solving for ω and A .

Final Algebraic Solution

So we now solve

$$6 = A \sin \alpha,$$

$$10 = A \sin(\omega + \alpha),$$

$$6 = A \sin(2\omega + \alpha).$$

Expand the second equation:

$$A \sin(\omega + \alpha) = A(\sin \alpha \cos \omega + \cos \alpha \sin \omega) = 10.$$

Using

$$A \sin \alpha = 6,$$

this becomes

$$6 \cos \omega + A \cos \alpha \sin \omega = 10. \tag{1}$$

Now expand the third equation:

$$A \sin(2\omega + \alpha) = A(\sin \alpha \cos 2\omega + \cos \alpha \sin 2\omega) = 6.$$

Again using

$$A \sin \alpha = 6,$$

we get

$$6 \cos 2\omega + A \cos \alpha \sin 2\omega = 6. \tag{2}$$

Using

$$\sin 2\omega = 2 \sin \omega \cos \omega,$$

equation (2) becomes

$$6 \cos 2\omega + 2A \cos \alpha \sin \omega \cos \omega = 6. \tag{3}$$

From equation (1),

$$A \cos \alpha \sin \omega = 10 - 6 \cos \omega.$$

Substitute this into equation (3):

$$6 \cos 2\omega + 2 \cos \omega (10 - 6 \cos \omega) = 6.$$

Now use

$$\cos 2\omega = 2 \cos^2 \omega - 1.$$

Then

$$6(2 \cos^2 \omega - 1) + 20 \cos \omega - 12 \cos^2 \omega = 6.$$

So

$$12 \cos^2 \omega - 6 + 20 \cos \omega - 12 \cos^2 \omega = 6,$$

which simplifies to

$$20 \cos \omega = 12.$$

Hence

$$\cos \omega = \frac{3}{5}.$$

Therefore,

$$\omega = \cos^{-1}\left(\frac{3}{5}\right).$$

Finally, since

$$\nu = \frac{\omega}{2\pi},$$

The frequency is

$$\nu = \frac{1}{2\pi} \cos^{-1}\left(\frac{3}{5}\right) \text{ Hz.}$$

Numerically,

$$\omega \approx 0.9273 \text{ rad s}^{-1}, \quad \nu \approx \frac{0.9273}{2\pi} \approx 0.148 \text{ Hz.}$$

So,

$$\nu \approx 0.148 \text{ Hz.}$$

Mechanical Energy in SHM

The total mechanical energy of a particle in SHM is the sum of its kinetic energy and potential energy:

$$E = K + U.$$

Kinetic Energy

The kinetic energy is

$$K = \frac{1}{2}mv^2.$$

Using

$$v = \pm\omega\sqrt{A^2 - x^2},$$

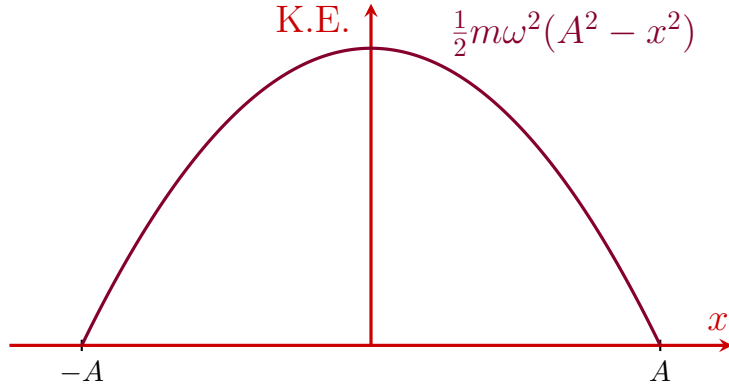
we get

$$K = \frac{1}{2}m\omega^2(A^2 - x^2).$$

Hence,

$$K = \frac{1}{2}m\omega^2(A^2 - x^2).$$

(6)



If

$$x = A \sin(\omega t + \alpha),$$

then

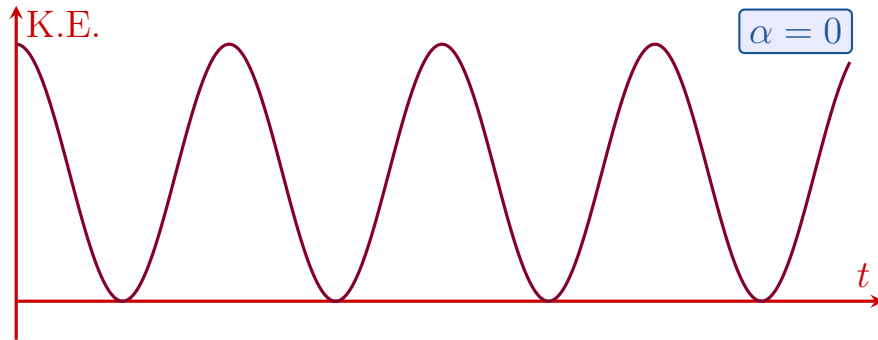
$$x^2 = A^2 \sin^2(\omega t + \alpha),$$

so

$$K = \frac{1}{2}m\omega^2 A^2 (1 - \sin^2(\omega t + \alpha)).$$

Thus,

$$\boxed{K = \frac{1}{2}m\omega^2 A^2 \cos^2(\omega t + \alpha).} \quad (7)$$



Potential Energy and Force

If a force $F(x)$ depends on position, then the corresponding potential energy $U(x)$ satisfies

$$\boxed{F(x) = -\frac{dU}{dx}.} \quad (8)$$

Physical Meaning

1. Every physical system has a tendency to move toward a state of lower potential energy.
2. The **force** is the agent that drives the system in that direction.
3. The derivative

$$\frac{dU}{dx}$$

tells us how rapidly the potential energy changes with position.

4. If

$$\frac{dU}{dx} > 0,$$

then potential energy increases as x increases. So the system should be pushed toward smaller x , where the potential energy is lower.

5. If

$$\frac{dU}{dx} < 0,$$

then potential energy decreases as x increases. So the system should be pushed toward larger x .

6. That is why the force must point in the direction opposite to the slope of the potential-energy curve:

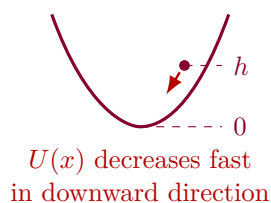
$$F(x) = -\frac{dU}{dx}.$$

7. The **magnitude** of the force is also connected to the steepness of the curve:

- if the potential-energy curve is steep, $\left|\frac{dU}{dx}\right|$ is large, so the force is large,
- if the curve is shallow, $\left|\frac{dU}{dx}\right|$ is small, so the force is small.

8. So, in words:

The force always acts in the direction in which the potential energy decreases, and it acts more strongly where the potential-energy curve is steeper.



A potential-energy curve can be read like a landscape: the particle tends to move “downhill,” and the steeper the downhill direction, the stronger the force.

Potential Energy in SHM

For SHM,

$$F(x) = -m\omega^2 x.$$

So

$$-\frac{dU}{dx} = -m\omega^2 x.$$

Hence

$$\frac{dU}{dx} = m\omega^2 x.$$

Integrating,

$$U(x) = \int m\omega^2 x \, dx = \frac{1}{2}m\omega^2 x^2 + C. \quad (9)$$

Potential energy always has an arbitrary reference level. Let us choose the equilibrium position $x = 0$ as the point where

$$U = 0.$$

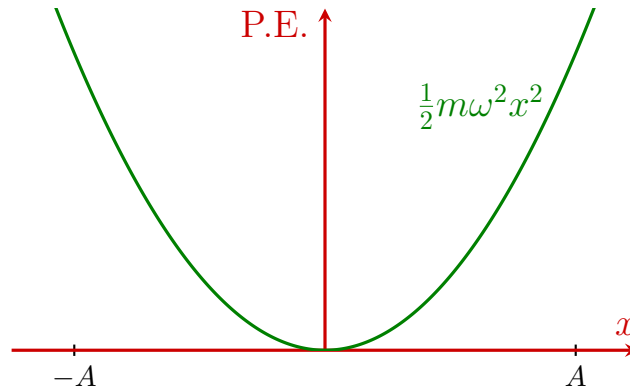
Then using eq. 9,

$$0 = \frac{1}{2}m\omega^2 \cdot (0)^2 + C$$

$$C = 0$$

Therefore

$$U(x) = \frac{1}{2}m\omega^2 x^2. \quad (10)$$

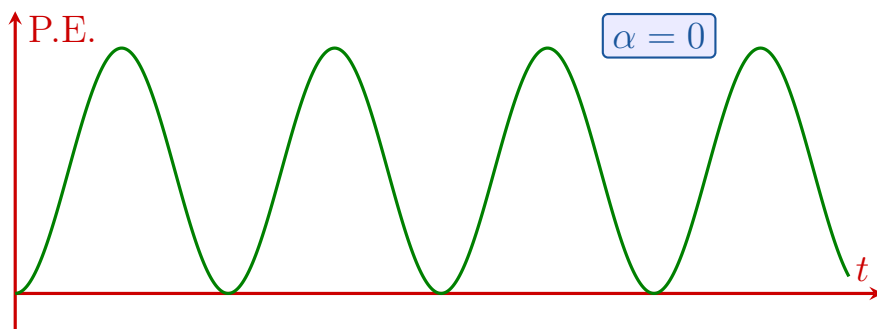


Substituting

$$x = A \sin(\omega t + \alpha),$$

we get

$$U(t) = \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \alpha). \quad (11)$$



Total Mechanical Energy

Now

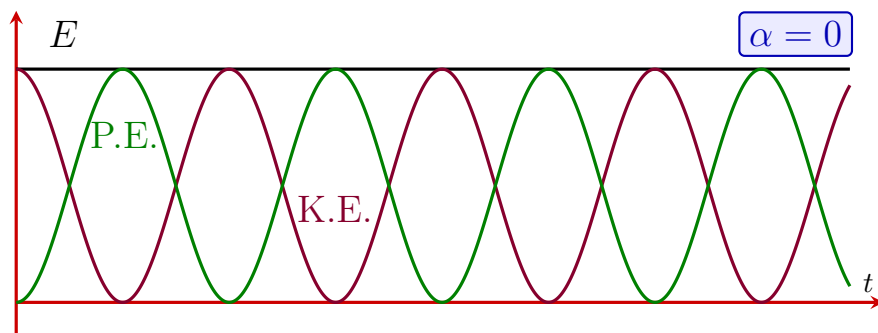
$$E = K + U = \frac{1}{2}m\omega^2(A^2 - x^2) + \frac{1}{2}m\omega^2 x^2.$$

So

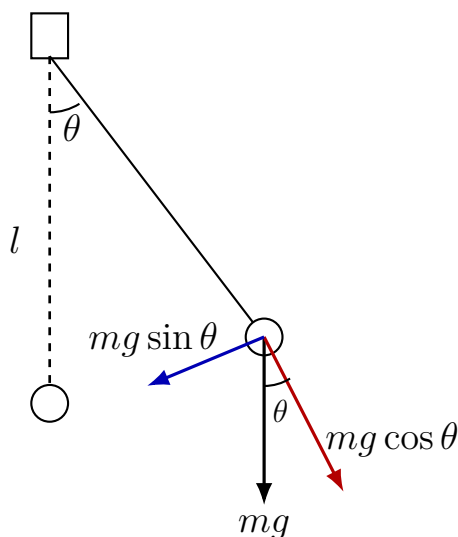
$$E = \frac{1}{2}m\omega^2 A^2. \quad (12)$$

This is independent of time.

In ideal SHM, the total mechanical energy remains constant. Kinetic and potential energies keep converting into one another, but their sum stays fixed. This is expected as there is no dissipation, so total energy should be conserved



Example: Simple Pendulum for Small Oscillations



For a simple pendulum of length l and bob mass m , the restoring force along the arc is

$$F_r = -mg \sin \theta.$$

For small angles,

$$\sin \theta \approx \theta.$$

So

$$F_r \approx -mg\theta.$$

Using

$$x = l\theta \quad \Rightarrow \quad \theta = \frac{x}{l},$$

we get

$$F_r = -mg \frac{x}{l}.$$

Comparing with

$$F = -kx,$$

we identify

$$k = \frac{mg}{l}.$$

Therefore,

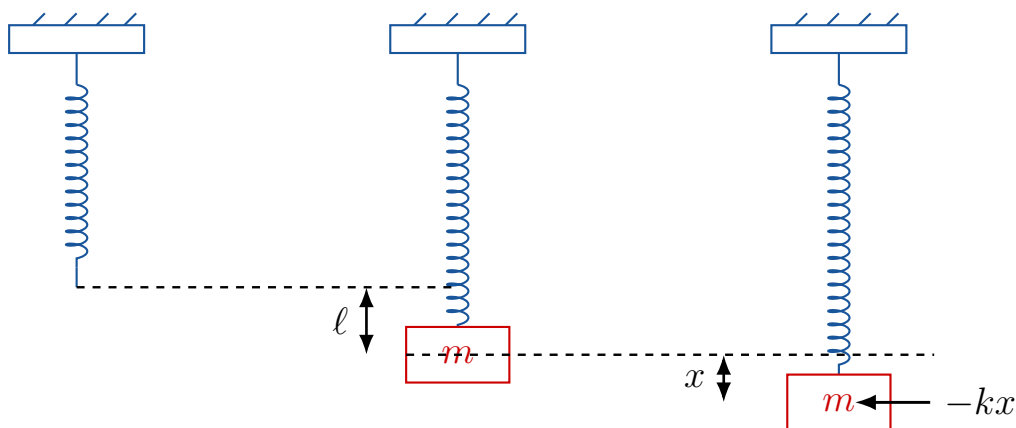
$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{l}}.$$

Hence the time period is

$$T = 2\pi\sqrt{\frac{l}{g}}.$$

For small oscillations of a simple pendulum, the time period depends on l and g , but not on the mass of the bob.

Example: Vertical Spring-Mass System



Consider a mass m attached to a vertical spring. Let the spring extend by an amount l under the weight of the mass before oscillation begins.

At equilibrium,

$$kl = mg.$$

So

$$k = \frac{mg}{l}.$$

Now, for oscillations about the new equilibrium position, the restoring force is

$$F = -kx.$$

Hence,

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{l}}.$$

Therefore,

$$T = 2\pi\sqrt{\frac{l}{g}}.$$

For the vertical spring, the l appearing in

$$T = 2\pi\sqrt{\frac{l}{g}}$$

is not the actual length of the spring. It is the extension produced by the attached mass in static equilibrium.

Final Perspective

This class develops the SHM story further by showing how:

- the time period and frequency arise from the phase of the sine function,
- displacement and velocity differ in phase by $\pi/2$,
- numerical questions can be solved directly from the standard SHM form,
- kinetic and potential energy exchange with time,
- and the total mechanical energy remains constant in ideal SHM.

It also shows how simple pendulum motion and vertical spring oscillation can be connected back to the same SHM framework.

Key Takeaway:

Simple harmonic motion is not only sinusoidal in form; it also has a beautiful internal structure: a fixed time period, a well-defined phase, and a constant total mechanical energy with continuous exchange between kinetic and potential parts.